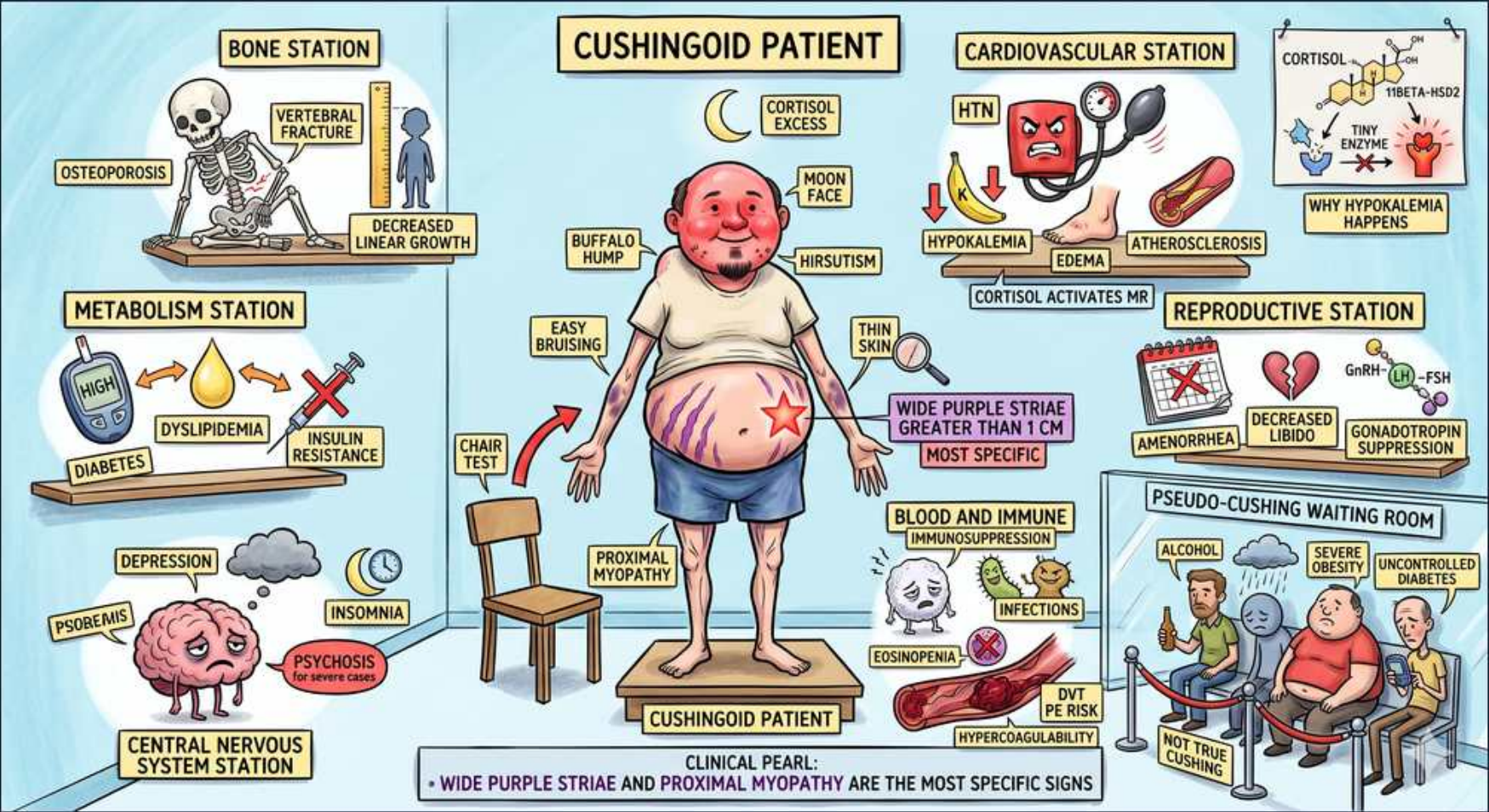




1. Central obesity + buffalo hump

WHAT YOU SEE

Bald cushingoid man drawn with a big round trunk, thin arms/legs, and a labelled 'BUFFALO HUMP' fat pad on the upper back — the central-fat-with-skinny-limbs body literally IS cortisol's fat-redistribution pattern.

WHAT IT ENCODES

Cortisol drives a characteristic FAT REDISTRIBUTION: central/truncal obesity with thin extremities ('lemon-on-matchsticks'), a dorsocervical fat pad (buffalo hump), and supraclavicular fat pads. Mechanism — cortisol promotes lipolysis in peripheral fat but lipogenesis centrally and stimulates appetite. The thin limbs reflect proximal muscle wasting from protein catabolism.

BOARD TRAP

Central obesity and buffalo hump are SENSITIVE but NOT specific — simple metabolic obesity looks similar. The specific signs are wide purple striae and proximal myopathy, not the fat distribution alone.

2. Moon face

WHAT YOU SEE

'MOON FACE' label, red round cheeks, 'CORTISOL EXCESS'

WHAT IT ENCODES

Moon facies (rounded, full face) plus facial PLETHORA (red cheeks from skin thinning and capillary visibility/polycythemia) are classic cortisol-excess findings. They develop gradually in endogenous Cushing — a slow change visible by comparing old photographs, a useful clinical clue.

BOARD TRAP

Moon face is sensitive but not specific (also seen in obesity, hypothyroidism, exogenous steroids); rely on the specific signs to confirm true Cushing.

3. Wide purple striae (most specific)

WHAT YOU SEE

Purple box 'WIDE PURPLE STRIAE GREATER THAN 1 CM — MOST SPECIFIC'

WHAT IT ENCODES

Wide (>1 cm) violaceous/purple striae are among the MOST SPECIFIC signs of Cushing. Cortisol's catabolism of collagen and dermal protein thins the skin so the underlying dermal vasculature shows through the stretched, torn tissue — hence the purple color and width. Located on abdomen, thighs, breasts, axillae.

BOARD TRAP

Thin, pale, narrow striae of ordinary obesity or pregnancy are NOT specific — only WIDE (>1 cm) PURPLE striae are. Don't accept pale striae as evidence of Cushing.

4. Proximal myopathy / chair test

WHAT YOU SEE

Empty chair, 'PROXIMAL MYOPATHY', 'CHAIR TEST'

WHAT IT ENCODES

Proximal muscle weakness and wasting (difficulty rising from a chair without using arms, climbing stairs, combing hair) is, with wide purple striae, one of the TWO MOST SPECIFIC discriminating signs. Mechanism — cortisol-driven proteolysis of type II muscle fibers. The 'chair test' (rise from a low chair with arms crossed) is the bedside screen.

BOARD TRAP

This is proximal (shoulder/hip girdle) weakness, NOT distal — distal weakness or sensory loss points to neuropathy, not Cushing. CK is usually NORMAL (steroid myopathy is catabolic, not inflammatory).

5. Easy bruising / thin skin

WHAT YOU SEE

'EASY BRUISING', 'THIN SKIN' labels

WHAT IT ENCODES

Cortisol's catabolic effect on collagen and dermal connective tissue produces thin, fragile, parchment-like skin that bruises with minimal trauma (ecchymoses) and heals poorly. Skin thickness <2 mm and spontaneous bruising are relatively SPECIFIC discriminating signs in younger patients.

BOARD TRAP

Easy bruising/thin skin is age-dependent — common in the elderly and on antiplatelet/anticoagulant therapy. It is most discriminating in a YOUNG patient where it would otherwise be unexpected.

6. Bone station

WHAT YOU SEE

Skeleton, 'OSTEOPOROSIS', 'VERTEBRAL FRACTURE', 'DECREASED LINEAR GROWTH'

WHAT IT ENCODES

Glucocorticoids cause OSTEOPOROSIS by suppressing osteoblasts, increasing osteoclast lifespan (RANKL), reducing intestinal calcium absorption and increasing renal calcium loss. Result: low bone density, fragility/vertebral compression fractures (often the presenting feature), and avascular necrosis. In CHILDREN, cortisol excess causes growth arrest/short stature — growth deceleration with weight gain in a child is a key Cushing clue.

BOARD TRAP

In a child, weight gain with NORMAL or accelerated linear growth points to simple obesity; weight gain with DECREASED linear growth (falling off the height curve) points to Cushing/endocrine cause.

7. Metabolism station

WHAT YOU SEE

Glucometer 'HIGH', 'DYSLIPIDEMIA', 'INSULIN RESISTANCE'

WHAT IT ENCODES

Cortisol is a counter-regulatory/diabetogenic hormone: it increases hepatic gluconeogenesis, induces peripheral insulin resistance, and impairs glucose uptake → impaired glucose tolerance or frank diabetes. It also causes dyslipidemia (high triglycerides/LDL) and contributes to the metabolic syndrome and central obesity. Hyperglycemia plus the Cushingoid phenotype is high-yield.

BOARD TRAP

New or worsening diabetes/HTN in a young, non-obese patient with Cushingoid features should prompt a Cushing workup rather than just being treated as primary type 2 diabetes.

8. Cardiovascular: HTN + hypokalemia

WHAT YOU SEE

BP cuff 'HTN', 'HYPOKALEMIA', 'EDEMA', 'CORTISOL ACTIVATES MR', 'ATHEROSCLEROSIS'

WHAT IT ENCODES

Hypertension is common via several mechanisms: cortisol increases vascular sensitivity to catecholamines/angiotensin, raises hepatic angiotensinogen, and — in high concentrations — overwhelms renal 11β -HSD2 so cortisol activates the MINERALOCORTICOID RECEPTOR → sodium/water retention, hypokalemia, edema. Accelerated atherosclerosis and a prothrombotic state make cardiovascular disease the leading cause of death in Cushing.

BOARD TRAP

Hypokalemic metabolic alkalosis is most pronounced in ECTOPIC ACTH/severe hypercortisolism (cortisol levels high enough to saturate 11β -HSD2) — its presence argues against a mild pituitary cause and toward ectopic ACTH or ACC.

9. CNS station

WHAT YOU SEE

Brain, 'DEPRESSION', 'INSOMNIA', 'PSYCHOSIS', 'PSYCHOSIS'

WHAT IT ENCODES

Neuropsychiatric effects are common and often underappreciated: depression (most frequent), anxiety, emotional lability, insomnia (loss of normal nocturnal cortisol nadir), impaired memory/cognition, and in severe cases mania or frank psychosis. Symptoms frequently improve after the hypercortisolism is corrected.

BOARD TRAP

Depression itself is a PSEUDO-Cushing state (it can mildly raise cortisol and cause a false-positive dexamethasone non-suppression), so don't assume every depressed patient with a borderline screen has true Cushing — confirm with multiple tests.

10. Reproductive station

WHAT YOU SEE

'AMENORRHEA', 'DECREASED LIBIDO', 'GnRH/LH/FSH SUPPRESSION', 'HIRSUTISM'

WHAT IT ENCODES

Excess cortisol suppresses the hypothalamic GnRH pulse generator → low LH/FSH → hypogonadism: oligo/amenorrhea, infertility, decreased libido, and erectile dysfunction. In women, concurrent adrenal androgen excess adds hirsutism and acne. These reproductive complaints are often the presenting symptom in young women.

BOARD TRAP

Marked hirsutism/virilization (deep voice, clitoromegaly) with Cushing signals strong ADRENAL ANDROGEN output — think adrenocortical CARCINOMA rather than a benign cortisol-only adenoma or simple pituitary disease.

11. Immune: infections + eosinopenia

WHAT YOU SEE

'IMMUNOSUPPRESSION', 'INFECTIONS', 'EOSINOPENIA', 'DVT PE RISK', 'HYPERCOAGULABILITY'

WHAT IT ENCODES

Cortisol is immunosuppressive: it impairs T-cell and neutrophil function and predisposes to bacterial, fungal, and opportunistic infections (which may present atypically/blunted). The CBC shows neutrophilia with LYMPHOPENIA and EOSINOPENIA (cortisol redistributes/apoptoses eosinophils and lymphocytes). Cushing is also a HYPERCOAGULABLE state (increased factor VIII, vWF, fibrinogen) → elevated VTE/DVT/PE risk.

BOARD TRAP

Cortisol causes eosinoPENIA and lymphoPENIA but neutroPHILIA (demargination) — don't expect the eosinophilia or leukopenia seen in adrenal INSUFFICIENCY; Cushing is the mirror image.

12. Pseudo-Cushing waiting room

WHAT YOU SEE

Waiting room: 'ALCOHOL', 'SEVERE OBESITY', 'UNCONTROLLED DIABETES', 'NOT TRUE CUSHING'

WHAT IT ENCODES

Pseudo-Cushing states cause genuine mild HPA-axis activation and overlapping features but are NOT true Cushing: chronic alcohol use, severe obesity/metabolic syndrome, poorly controlled diabetes, major depression, and physiologic stress/critical illness. They can yield false-positive screening tests. Distinguishing tools: dexamethasone-CRH test and midnight cortisol; removing the trigger (e.g., alcohol abstinence) normalizes the axis.

BOARD TRAP

These mimics produce FALSE-POSITIVE screens — which is exactly why confirmation requires ≥ 2 abnormal first-line tests and, when ambiguous, a dex-CRH test. Don't diagnose Cushing off a single borderline result in an alcoholic, obese, or depressed patient.